

On the Use of Requirements Measures to Predict Software Project and Product Measures in the Context of Android Mobile Apps: a Preliminary Study

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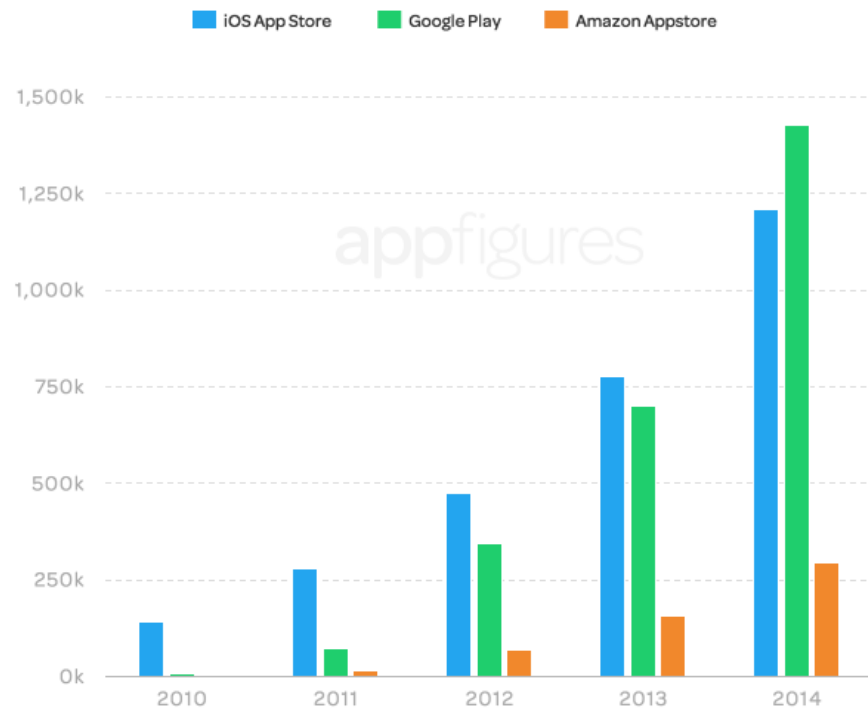
Outline

- Motivation and Proposal
- Empirical study
 - Design
 - Results
- Conclusions and future work

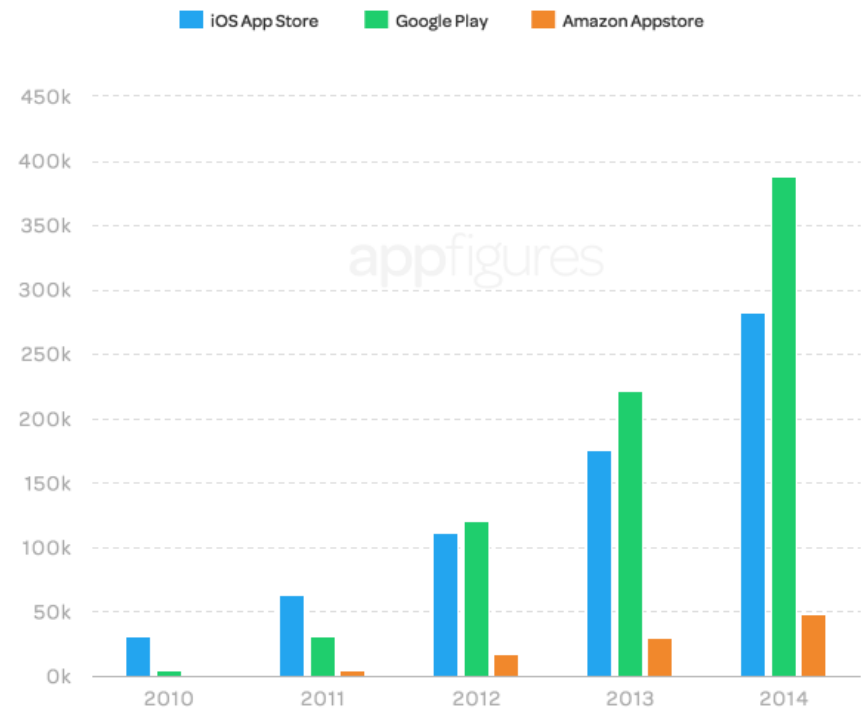
Motivation and Proposal

Apps and Developers Growth

Total Number of Apps by App Store



Total Number of Developers by App Store



Mobile Apps and Software Engineering

- **Software Engineering** processes can be employed to improve the quality of these applications
- one option is **software estimation**
 - used into project plans, iteration plans, budgets, investment analyses, pricing processes, bidding rounds ...
- Approaches: (i) expert estimations, (ii) formal estimation models, (iii) a combination-based model
- there is not **the best approach**

Mobile Apps and Software Engineering (2)

- Estimation accuracy can depend on the software project contexts and technologies employed for the development
 - an approach useful for desktop applications, useless for web applications and mobile apps
- For formal estimation models, differences are also due to the software artifacts employed
 - models built in the early phase are more useful than those built later (e.g., exploiting LOC) for estimation purposes
 - but less accurate

Proposal

- Compare software **project** and **product** measures in the development of mobile apps for Android devices
- Build **formal estimation models** to predict Effort and Graphical components
- Exploit information from **requirements documents**
 - use cases, actors, classes, ...

Proposal (2)

- Resources: Smartphones are equipped by sensors (e.g., accelerometer, gyroscope, GPS, brightness and temperature) and offer communication features, including phone calls, SMS, email, and camera functionalities
- Activity: an app component that provides a screen with which users can interact to do something (e.g., dial the phone, take a photo, send an email, or view a map)
- Service: a background component that performs either long-running operations or works for remote processes - No GUI - (e.g., fetch data over the network)

Empirical Study

Goal and Research questions

- **Analyze** the use of *measures* obtained from *requirements and analysis document* **for the purpose of** building *prediction models with respect to* the *development effort* and the *number of graphical components* of a GUI mobile app **from the point of view of** the *researchers* and the *professional* developers **in the context of** mobile apps developed by *young developers* grouped in *small teams*
- RQ1: Do the measures obtained from the requirements and analysis document provide accurate predictions of effort of mobile applications?
- RQ2: Do the measures obtained from the requirements and analysis document provide accurate predictions of the number of graphical components of mobile applications?

The baseline

- Predictions achieved with software measures from source code
- Information collected with Understand
- Rationale: measures obtained early have to not perform significantly worse than measures obtained later in the development process

Context

- 46 Bachelor students in Computer Science at the University of Salerno, grouped in 23 teams
- Use of GitHub for the management activities
- Scheduled activities: deadlines and deliverables
 - Project proposal, Requirements Analysis Document (RAD), Intermediate releases (3), Final release
- Students followed an incremental prototyping development approach. Adoption of pair programming also promoted

Context (2)

- A pre-questionnaire to collect some demographic information and development experiences
 - students were generally smart-users for mobile apps, they did not know Android development, most of them good Java developers, with database development experience
 - 30 have experience with software engineering practices to develop desktop and web applications
 - 14 have good network programming experience

Context (3)

- A market analysis also conducted and presented to the lecturer
- Imposed requirements:
 - the app had to interact with a remote server, through JSON
 - the app had to exploit native device functionalities, including maps, GPS, phone call and SMS
 - the app had to manage the device rotation, and use SQLite
- Approved apps types: game, social networking, geolocalized services, sports, entertainment, travel, restaurant, transport, public administration, and people management

Variables

- Information **from RAD** to predict the effort and graphical components
 - conjecture: e.g., effort is proportional (on a first approximation) to the number of use cases, classes, etc.

Dependent variables

Measure	Description
Effort	The total effort to develop a mobile app expressed in terms of person/hour
XMI_UI	Number of XMI files about graphical elements (e.g., text box and so on) of the mobile app

Independent variables

Measure	Description
FR	Number of Functional Requirements
Act	Number of Use Case Actors
UC	Number of Use Cases
Cla	Number of Classes
SD	Number of Sequence Diagrams

Measure	Description (from source code)
McB	McCabe Cyclomatic complexity
Classes	Number of Classes
Files	Number of Files
Methods	Number of Methods
NL	Number of all Lines
LOC	Number of Lines of Source Code
CLOC	Number of Lines of Comment
STM	Number of Statements

Baseline

Data Analysis: estimation technique

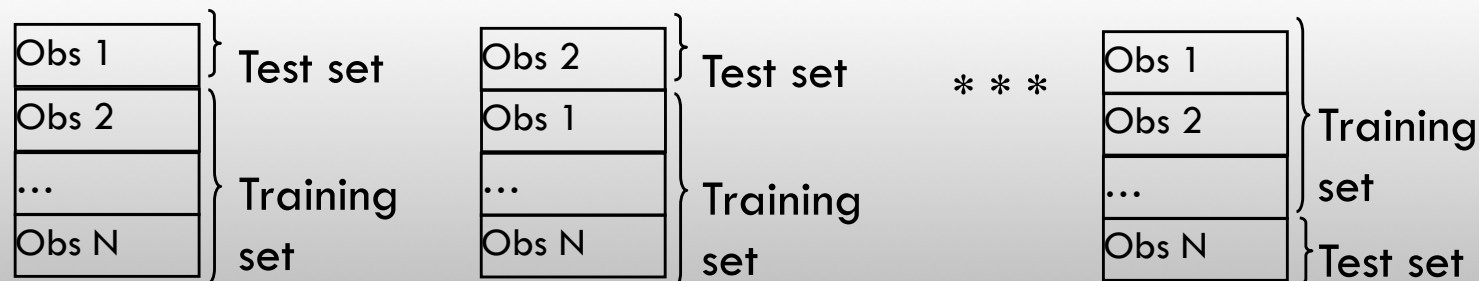
- StepWise Linear Regression (SWLR)

$$y = b_1 * x_1 + b_2 * x_2 + \dots + b_n * x_n + c$$

- y : dependent variable (i.e., Effort, XMI_UI)
- x_i : independent variable (e.g., FR, Act, UC, Cla, SD)
- SWLR computes equation in stages, according to:
 - (forward) **Stage 0**: No variables in; **Stage i**: includes the variables one by one if they are statistically significant correlated with the dependent variable
 - (backward) **Stage 0**: All the variables in; **Stage i**: removes the variables one by one if they are not statistically significant correlated with the dependent variable
 - a combination of forward and backward

Data Analysis: prediction validation

- We verified whether or not the obtained prediction values are useful estimations of the actual values
- We carried out a *leave-one-out* (LOO) cross validation
 - the original data set is divided into N=23 different subsets of training and validation sets
 - training set to build the estimation model and test set to validate the obtained prediction



Data Analysis: evaluation criteria

- *Mean of AR* (MAR) *Median of AR* (MdAR)
- AR (Absolute Residuals)

$$| \text{Effort}_{\text{actual}} - \text{Effort}_{\text{predicted}} |$$

$$| \text{XMI_UI}_{\text{actual}} - \text{XMI_UI}_{\text{predicted}} |$$

- Wilcoxon test to verify whether there was statistically significant difference between the absolute residuals achieved with the models based on RAD and Source Code (SC) measures
 - we accepted a probability of 5% of committing a Type-I-Error

Threats to Validity

- External validity:
 - mobile apps from a specific context (replications are needed)
 - complexity and size of apps are not far from those of a market place
 - use of students as developers (apps often developed by young professionals and it is difficult to find people skilled on these technologies)
- Conclusion validity:
 - evaluation criteria can be considered objective. Statistical tests were properly applied verifying hypothesis
 - the number of apps composing the data set (replications are needed)

Threats to Validity

- Internal validity:
 - The treatments or the experiences of the participants might also affect this kind of validity
 - widely used guidelines to control all the possible extraneous factors
- Construct validity:
 - other independent variables could be considered (future work direction)

Results and Discussion

- We verified the linear regression assumptions
 - a log transformation of the variables
- Analysis of outliers: Cook's distance and stability analysis to verify influential observations

Model based on RAD measures

Dependent Var.	Independent Var.
Effort	Act, Cla
XMI_GUI	UC, Cla

Model based on SC measures

Dependent Var.	Independent Var.
Effort	Classes, NL
XMI_GUI	McB, LOC, STM, DIT

Results and Discussion

- We verified the linear regression assumptions
 - a log transformation of the variables
- Analysis of outliers: Cook's distance and leverage analysis to verify influential observations

Plausible justification: the number of classes in a requirements specification document represents the basis for the next phases of the development process (i.e., a developer uses these classes as the starting point for development)

Model based on RAL

Features

Dependent Var.	Independent	Dependent Var.	Independent
Effort	Act, Cla	Effort	Classes, NL
XMI_GUI	UC, Cla	XMI_GUI	McB, LOC, STM, DIT

Results and Discussion

- We verified the linear regression assumptions
 - a log transformation of the variables
- Analysis of outliers: Cook's distance and stability analysis to verify influential observations

Model based on R² and F-statistics

Dependent Var.	Independent Var.	Dependent Var.	Independent Var.
Effort	Act, Cla	Effort	Classes, NL
XMI_GUI	UC, Cla	XMI_GUI	McB, LOC, STM, DIT

Plausible justification: in mobile app development there is a relationship between use cases and GUIs since mobile apps are mainly devised to present data rather than to perform their manipulation

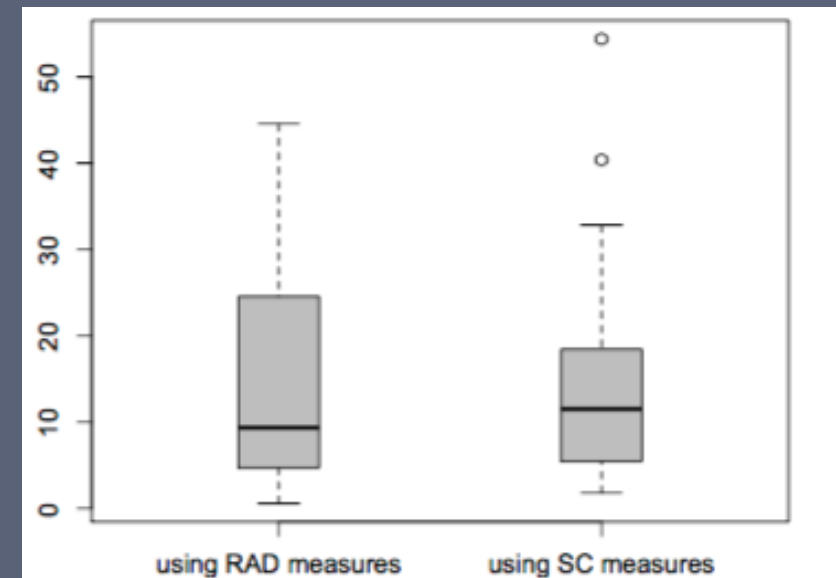
Results and Discussion

(RQ1)

Predicting Effort

- **MAR** values are quite close
- **MdAR** value of RAD measures is even better
- No statistically significant difference in the AR values (wilcoxon test p-value = 0.985)
- *Measures obtained from the requirements and analysis document provide accurate predictions of effort of mobile applications*

Model	MAR	MdAR
RAD measures	15.63	9.37
SC measures	14.34	11.55



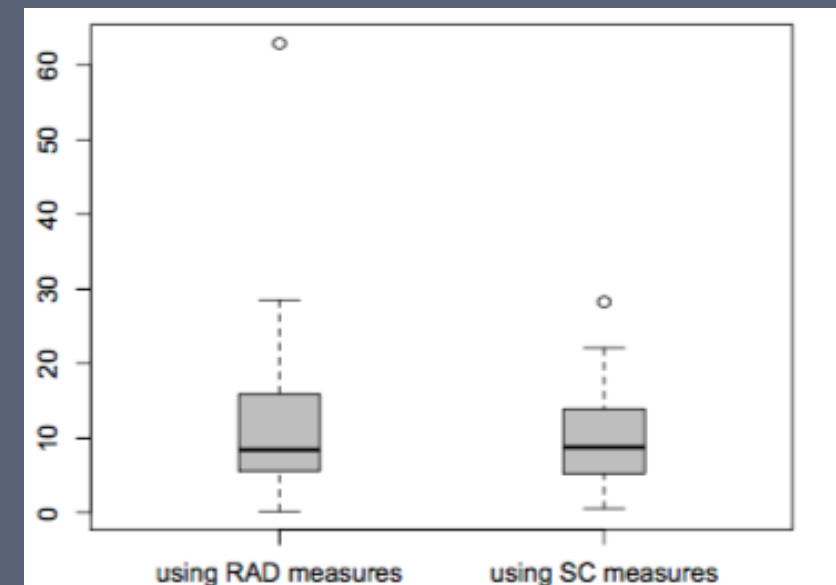
Results and Discussion

(RQ2)

Predicting XMI_UI

- **MAR** value of SC measures slightly better
- **MdAR** values very close
- No statistically significant difference in the AR values (wilcoxon test p-value = 0.773)
- *Measures obtained from the requirements and analysis document provide accurate predictions of graphical components of mobile applications*

Model	MAR	MdAR
RAD measures	12.15	8.41
SC measures	9.81	8.76



Conclusions and Future Work

Conclusions

- Software estimation in the context of mobile development has been marginally investigated
 - a systematic literature review to investigate applicability of traditional estimation models (by Souza and Aquino)
 - Analogy based estimation approaches for mobile apps, to predict size and cost (by Nitze et al.)
 - Functional size measurements of mobile apps, in terms of IFPUG and COSMIC, to estimate source code size (by van Heerengen and van Gorp, Ferrucci et al.)
- Differently from previous work, we focus on software project and product measures to estimate the effort needed to develop mobile apps as well as the number graphical components

Future Work

- Replications as revealed in Threats to validity discussion
 - larger datasets
 - different application domains
- Gather data in software projects that simulate in a different way how mobile apps are designed and developed (e.g., applying Agile methodologies)
 - definition of new project measures (e.g., size of the stories) and further replications

Questions? Comments?

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